# Themes in dynamics in dimension one and two

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## Examples in small dimension

Renormalisation:

- Rotations on $\mathbb{S}^1$;
- Interval exchange map;
- Homeomorphisms of $\mathbb{S}^1$ diffeomorphisms;
- The quadratic family $f_\lambda(x) = \underset{C}{\lambda}\overset{z}{x}(1-\overset{z}{x}), \underset{c\in\mathbb{C}}{\lambda \in \mathbb{R}^+}$ (or Mandelbrot family for the parameters in red);
- The Hénon family.

Defining $x_n$ inductively by taking $x_0 \in \mathbb{R}$ and $x_{n+1} = f_\lambda(x_n)$, we will study operators $R$ acting on dynamical systems from $\mathbb{R}$ to $\mathbb{R}$. Looking at the dynamics of this operator allows to deduce the dynamics of $f_\lambda$.

# 1 Rotations of the circle

Consider $\alpha \in \mathbb{R}$, the circle $\mathbb{S}^1 = \mathbb{R}/\mathbb{Z}$ and $R_\alpha : x \mapsto x + \alpha \mod \mathbb{Z}$. Our first goal is to show the 3 gaps

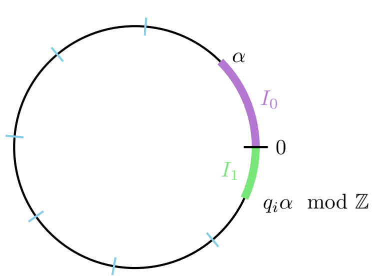
**Theorem 1.1 — 3 longueurs** For all $n \in \mathbb{N}$, $x \in \mathbb{S}^1$, the set $\mathbb{S}^1 \setminus \left\{R_\alpha^k(x)\right\}_{k\in[\![0,n-1]\!]}$ is composed of intervalls of at most 3 different lengths.

This was conjectured by Steinhaus, and solved in the 50s.
Let's analyze the orbits of $R_\alpha$.

**R** It is sufficient to take $x = 0$, to consider all possible orbits.

Suppose that $\alpha \in [0, \frac{1}{2}]$. The return map on $I_0 \cup I_1$ is given by:

$$x \in S^1 \mapsto \tau(x) = \min\{n \in \mathbb{N}^* \mid R_\alpha^n(x) \in I_0 \cup I_1\}$$

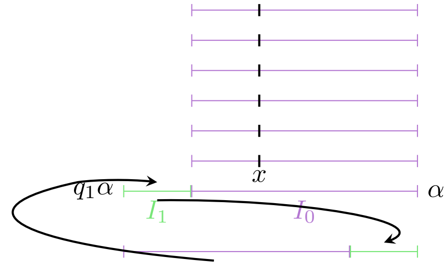



$T$ return map: $T(x) = R_\alpha^{\tau(x)}(x)$

**Lemma 1.2** If we glue the ends of $I_0 \cup I_1$, $T$ is a rotation.

$T$ acts as an exchange of 2 intervals: this is a rotation on $I_0 \cup I_1$ "glued"

Its angle is then $\frac{|I_0|}{|I_0|+|I_1|} = \frac{\alpha}{1+\alpha-q_1\alpha}$.




$q_{n+1} = \min\{q > q_n \mid \|q_n\alpha\| > \|q\alpha\|\}$ with for $x \in S^1$, $\|x\| = d(x, \mathbb{Z}) \in [0, \frac{1}{2}]$